Assignment 1: Data Detective

Introduction to Data Science · 2.5 course marks (marked out of 10) · Set in Week 4 · Due at the start of the Week 5 class

Deadline: 30th November (After this date, NO ASSIGNMENT WILL BE ACCEPTED)

Note: You can either push it to your own repository and send me the link on WhatsApp or send me a ZIP file via email (a6dulrauf@gmail.com)

The situation

You are a junior analyst at a small online shop. The owner sends you `orders.csv`, the same file you profiled in Week 2, and says: "I want to use this for next month's report. Can I trust it?" You will find out, fix what can be fixed honestly, and leave a record that anybody can follow.

The data

`orders.csv`: 200 orders, columns `order_id, customer_id, customer_ref, city, category, quantity, unit_price, order_date, status, notes`. Use the copy you used in Week 2, or the Week 2 data page, or the **orders.csv** dataset that is given. Put it in `data/raw/orders.csv` and never change it.

What you must do

1. **Profile first.** Before changing anything, write down what you find: shape, missing values per column, distinct values, odd values, formats. Use the six quality dimensions (completeness, validity, consistency, uniqueness, accuracy, timeliness) to name each problem.
2. **Write a cleaning script** that reads `data/raw/orders.csv` and writes `data/processed/orders_clean.csv`. All changes must be made in code. The raw file must stay exactly as it was.
3. **Write a decision log** with at least **8 rows**. Each row has four parts: *Problem*, *Evidence*, *Decision* and *Reason*. A reason that only says "because it was wrong" earns no credit.
4. **Write at least 6 data tests** with `assert`, and run them on the cleaned data. The tests must pass.
5. **Write a before-and-after summary:** rows before and after, missing values per column before and after, and any flag columns you added.
6. **Write half a page of questions for the shop owner:** the things you could not decide from the file alone.

Rules

- Never guess a value. If the correct value cannot be known from the file, set it to missing and say so in the log.
- Do not delete a row without a recorded reason.
- Flag unusual values (for example possible duplicates or outliers) instead of deleting them, unless they are impossible.

- Your notebook or script must run from top to bottom after a restart.

What to submit

A link to your Git repository (or a Colab notebook) containing: the script, the processed file, `decision_log.md` (or a table in the notebook), your tests and the half-page of questions. Add a short README that says how to run it.

How it is marked (10 marks)

Area	Marks	What earns full marks
Diagnosis of the problems	2	Finds most of the planted problems and names the quality dimension of each
Script and raw data protection	2	Runs from the top; raw file untouched; every change in code; clean structure
Decision log	3	At least 8 rows; evidence cited; the reason names the alternative that was rejected
Data tests	2	At least 6 meaningful asserts that pass, including a range check and a category check
Summary and questions for the owner	1	Honest before-and-after numbers; specific questions

Marks are reduced for invented values, for deleted rows with no reason, and for a notebook that cannot be rerun. Rewards go to reasoning, not to the number of rows removed.

Use of AI tools

You may use an AI assistant as a copilot. Attach your prompt log, and check every line of generated code against the data. You remain responsible for every decision in your log.

A hint about where to look

A good profile of this file will show: values that stand for “missing” in more than one way; a number column that is not entirely numeric; the same city written in several ways; rows that are shorter or longer than the header; a column that never changes; and two columns that carry the same information. Finding all of them is part of the job.